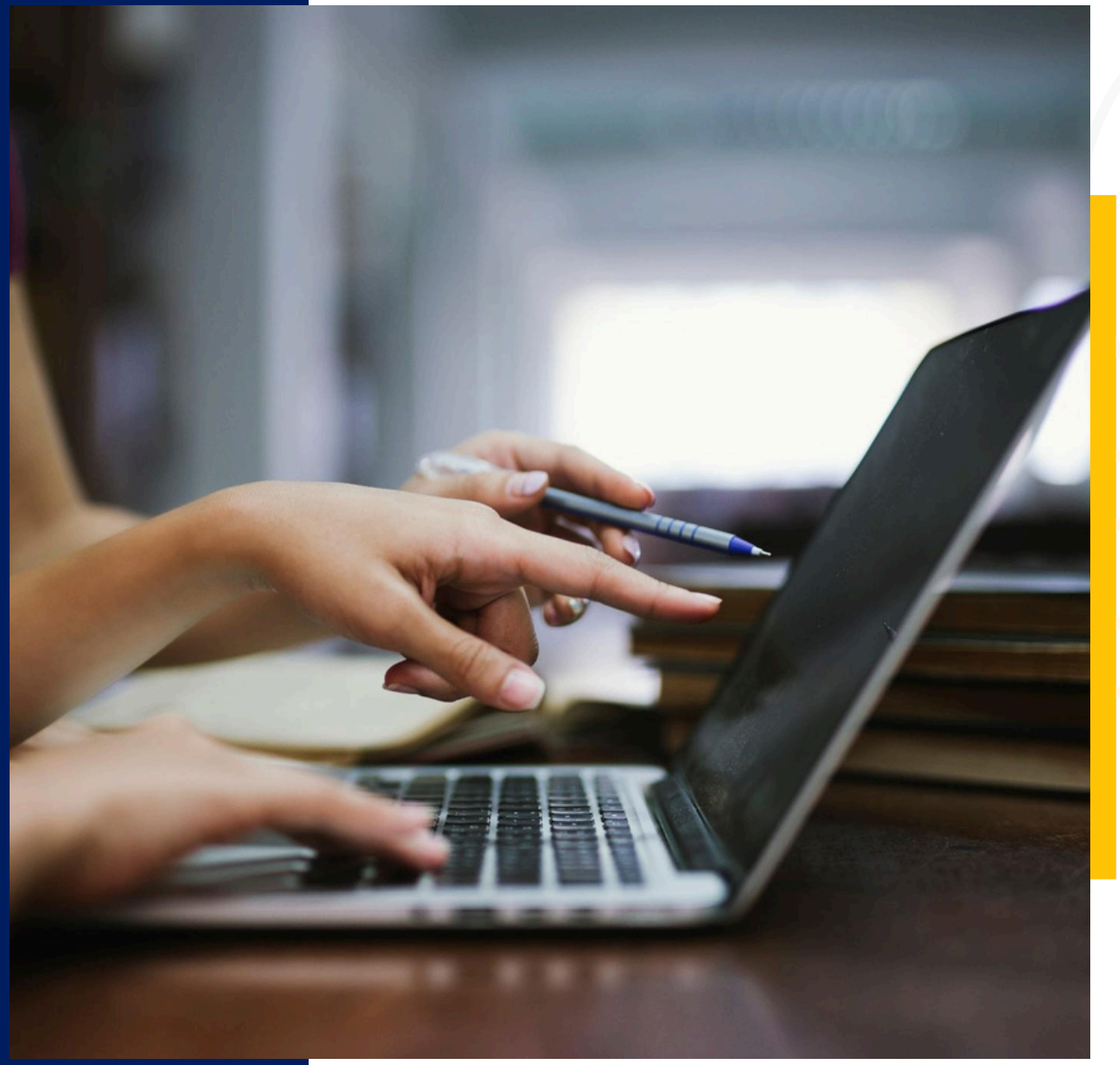

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Project Title : Smart AI Surveillance & Alert System using YOLO

Team ID: *CU_CP_Team_12315*

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Project Objectives

1. Problem Statement
2. Project Overview – Introduction
3. System Features
4. End Users
5. Dashboard & Alerts
6. Working Flow
7. Results
8. Conclusion
9. Results

Problem Statement

- Traditional CCTV systems only record footage
- No real-time intelligence or alerts
- Requires manual monitoring (inefficient)
- Important events may be missed
- No centralized dashboard system
- No automatic detection of objects/events
- Need for an AI-based automated surveillance system

Project overview -Introduction

AI-based surveillance system using YOLOv4
Uses deep learning to detect and identify objects in real-time

Dashboard to monitor multiple cameras
Provides a centralized interface to view and manage all camera feeds

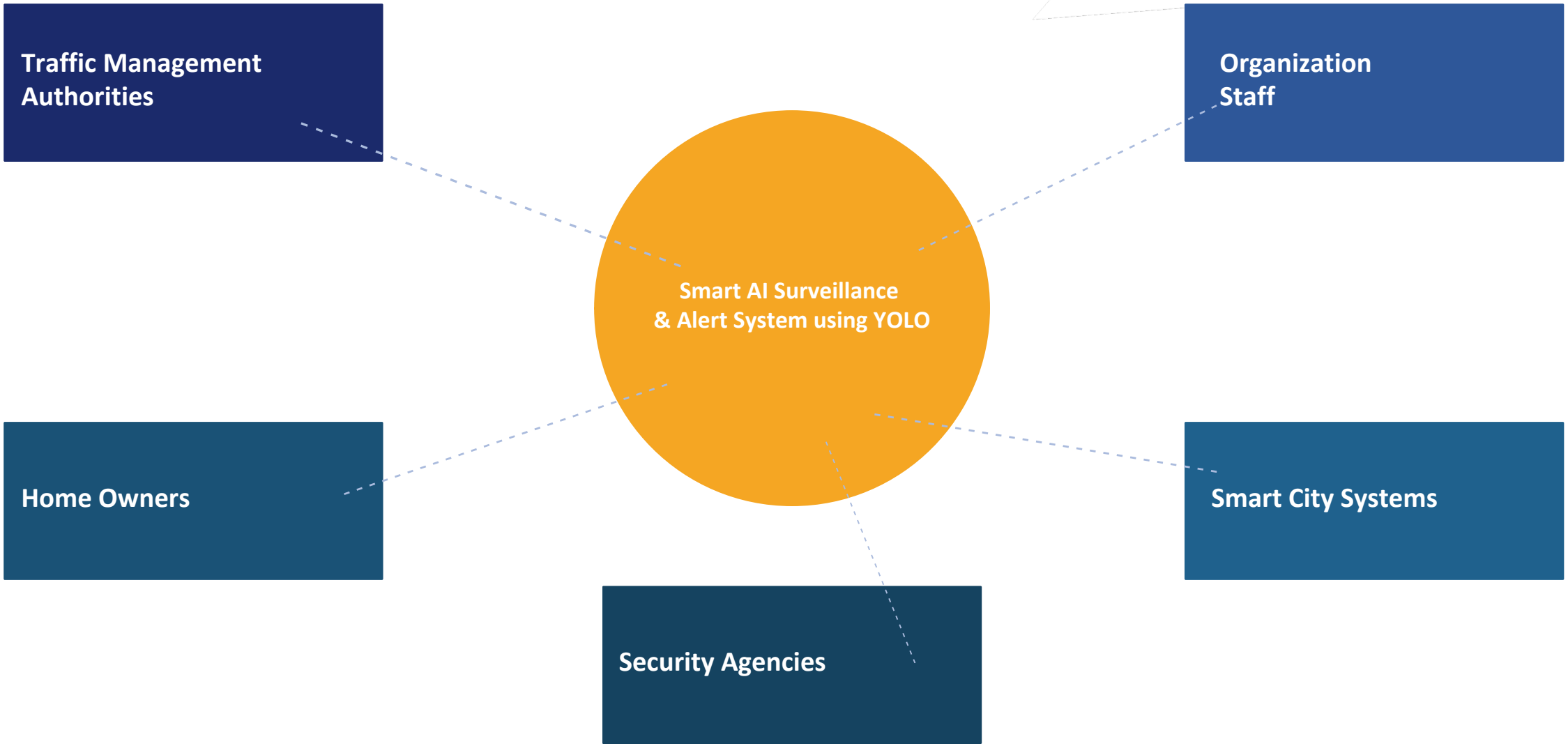
Detects objects in real-time
Continuously processes video frames to identify multiple objects instantly

Generates alerts for suspicious activity
Sends notifications when specific objects or conditions are detected

Supports CCTV camera integration
Connects directly to IP cameras for live video monitoring

Works for traffic monitoring systems
Helps in detecting vehicles, congestion, and violations, home security

End Users


Traffic Management Authorities
Organization Staff
Smart AI Surveillance & Alert System using YOLO
Home Owners
Smart City Systems
Security Agencies

Wow factor

Real-time object detection using YOLO

Instantly detects and labels objects from live video streams

Works on live CCTV feeds

Processes continuous video input directly from connected cameras

Smart dashboard interface

Displays detections, camera feeds, and alerts in a single view

Automatic alert system

Triggers notifications when predefined conditions are met

Multi-camera support

Can monitor and analyze multiple CCTV streams simultaneously

Multi-camera support

Flexible design suitable for both public and private environments

Modelling

1. CCTV Camera Input (RTSP/IP Camera)
2. Video Frames Captured
3. YOLO Model Processing
4. Object Detection
5. Alert Generation (if needed)
6. Data sent to Dashboard
7. User monitors in real-time

Result / Outcomes

90%

Real-Time Detection

0

Missed Events

100%

Alert Accuracy

↑

User Efficiency

↓

Human Errors

24/7

System Monitoring

Conclusion

AI improves surveillance efficiency

Reduces manual monitoring

Real-time detection enhances safety

Faster response to events

Smart alerts improve awareness

Instant notifications

Centralized dashboard simplifies control

All cameras in one place

System is scalable and flexible

Works for multiple use cases

Future Perspective

AI Behavior Detection

Detects unusual or suspicious activities automatically in real-time

Mobile App Integration

Allows users to monitor cameras and alerts from anywhere

Smart Alerts (SMS/Email)

Sends instant notifications for important events and detections

Advanced Analytics Dashboard

Provides insights on traffic patterns and activity trends

Face Recognition System

Identifies known individuals and detects unknown persons

Cloud-Based Monitoring

Enables remote access and centralized system management

Thank You